

In many sectors in Mathematics, the 'Invariant' objects are quite important.

Def. Let K/F be a finite separable extension, and let L/F be Galois Closure.

Define $N_{K/F}: K \rightarrow F; \alpha \mapsto \prod_{\sigma \in \text{Emb}_F(K, L)} \sigma(\alpha)$

$$\text{Tr}_{K/F}: K \rightarrow F; \alpha \mapsto \sum_{\sigma \in \text{Emb}_F(K, L)} \sigma(\alpha).$$

In particular, if K/F is Galois, then $L=K$, so that $\text{Emb}_F(K/L) = \text{Gal}(K/F)$.

Prop.

- 1). $N_{K/F}$ and $\text{Tr}_{K/F}$ are well-defined. (That is, for any $\alpha \in K$, $N_{K/F}(\alpha), \text{Tr}_{K/F}(\alpha) \in F$)
- 2). The $N_{K/F}$ Preserves multiplication and the $\text{Tr}_{K/F}$ Preserves addition.